

One Hundred Years of The Color Line: A Statistical Record of Baseball's Racial Barrier

Abstract

We construct a merged statistical record of professional baseball in the United States spanning 1871 to 2016, combining the Lahman Baseball Database (MLB), Retrosheet Negro Leagues game records (107,000 games, 1903-1962), and FiveThirtyEight Negro Leagues player ratings (480 players with WAR estimates). Using a six-stage computational pipeline, we report four findings. First, the integration of Negro Leagues statistics into MLB's record book — ratified by Major League Baseball in May 2024 — produces 90 rank changes across the top-20 career and single-season leaderboards in five offensive categories, with Josh Gibson displacing Babe Ruth atop the career OPS and slugging rankings. Second, the integration of the major leagues themselves, from Jackie Robinson's debut on April 15, 1947, to Pumpsie Green's on July 21, 1959, took 12 years, 3 months, and 6 days, with three franchises holding out past the midpoint. Third, we estimate the aggregate cost of the barrier at 4,472 lost player-seasons and 2,063.5 lost WAR across 453 careers, a figure roughly equivalent to erasing ten Hall of Fame careers from the record. Fourth, counterfactual career projections using quadratic WAR-by-age models suggest that Jackie Robinson, who debuted at 28, would have accumulated approximately 45.6 career WAR had he entered the majors at 21 — and that Roy Campanella, who debuted at 26, lost an estimated six seasons and 2.7 WAR to the barrier before a car accident ended his career at 36. The pipeline, derived datasets, and an interactive dashboard are open.

1. Introduction

On June 16, 2020, Major League Baseball announced that it would officially recognize the Negro Leagues as major leagues, retroactively granting major-league status to approximately 3,400 players whose careers had been statistically invisible for more than half a century. Four years later, in May 2024, MLB completed the integration of Negro Leagues statistics into its official record book.

The effect was immediate and sweeping. Josh Gibson — a catcher who played from 1930 to 1946 and died three months before Jackie Robinson broke the color line — rose to the top of baseball's career batting average, slugging percentage, and OPS leaderboards. Ninety rank changes rippled through the top-20 lists across five major statistical categories. Names that had existed in a parallel record for decades were suddenly, officially, part of the game's central narrative.

This paper examines those numbers and the barrier that kept them separate. We do not argue that statistics capture the full weight of what segregation meant for the men who lived it. We do argue that the numbers, assembled carefully and presented honestly, reveal the shape and cost of the barrier with a precision that narrative history alone cannot.

1.1 Two readings, one record

Two interpretations are available for every finding we report, and both are correct.

Under a **reading of exclusion and loss**, the data tells a story of systematic erasure. Four hundred and fifty-three players had their careers confined to a parallel league with shorter seasons, lower pay, and no access to the infrastructure that preserved MLB statistics for posterity. The aggregate cost — 4,472 lost seasons, 2,063.5 lost WAR — is a measure of talent that the major leagues refused. Josh Gibson died without playing a single major-league game. Oscar Charleston's entire career predates integration. Satchel Paige pitched twenty seasons in the Negro Leagues before getting a chance in the majors — at age 42. Under this reading, the record book, even after the 2024 merger, is an artifact of injustice.

Under a **reading of excellence and resilience**, the same data tells a story of extraordinary achievement under extraordinary constraints. Gibson's .372 career batting average was compiled in a league with roughly 60-game seasons, meaning he sustained that level of performance with less rest, worse travel, and fewer at-bats to smooth out variance. The Negro Leagues produced talent at a density that the merged leaderboards now make visible: 16 new entries in the career batting average top 20, 13 in career OPS. The men who played in these leagues were not diminished by segregation. They were the best players in the history of the game, playing in the only league that would have them. Under this reading, the record book is an overdue act of recognition.

Both readings predict the same statistical signatures, because both are simultaneously true. The barrier was real and the excellence was real and neither cancels the other. We return to this tension throughout.

2. The Data

Sources

We draw from four primary sources:

The Lahman Baseball Database provides comprehensive MLB statistics from 1871 to 2016 — batting, pitching, fielding, and biographical data for every player who appeared in a major-league game. This is our baseline record.

Retrosheet Negro Leagues contributes approximately 107,000 game records from 1903 to 1962, compiled from box scores, newspaper accounts, and archival research. These records vary in completeness: some games have full box scores, others have only line scores or game results.

FiveThirtyEight Negro Leagues Player Ratings provide WAR estimates and skill ratings for 480 Negro Leagues players, derived from available box scores and contemporary scouting assessments. These ratings were developed as part of FiveThirtyEight’s historical baseball analysis project and represent the most comprehensive quantitative assessment of Negro Leagues talent available.

The Seamheads Negro Leagues Database provides the statistical foundation that MLB used for its 2024 record-book integration, including the official batting lines that now appear in the merged leaderboards.

Library of Congress Photographs provide visual documentation of Negro Leagues players and games, used in the accompanying dashboard but not in the statistical analysis.

The threshold problem

Negro Leagues seasons were short — roughly 60 games compared to MLB’s 154 (later 162). This creates an immediate methodological question: what minimum thresholds should govern inclusion in career and season leaderboards?

MLB uses 200 at-bats for single-season rate statistics and approximately 2,000 career at-bats for career rate statistics. Applying these thresholds to Negro Leagues players would exclude almost everyone: a 60-game season might produce 200-250 at-bats for a full-time player, but many players fell short of even that.

We follow the Seamheads methodology adopted by MLB: 80 at-bats for single-season qualification and 500 career at-bats for career qualification. These thresholds are proportional to the season length and broadly equivalent in selectivity to the MLB thresholds. They are not identical, and this difference should be kept in mind when interpreting rate-statistic comparisons. A player sustaining a .370 batting average over 500 at-bats is not the same sample as sustaining it over 8,000, and the smaller sample carries larger confidence intervals.

This is a genuine limitation, not a technicality. We flag it here and return to it in our limitations section.

3. The Shape of the Data

The barrier in time

The color line in professional baseball hardened around 1900. Moses Fleetwood Walker, the last Black player in the major leagues of the 19th century, was forced out of the International League in 1889. By the turn of the century, the barrier was absolute: no written rule, no official policy, but a complete and enforced exclusion that held for 47 years.

The Negro National League was founded in 1920. The Eastern Colored League followed in 1923. For the next quarter-century, Black baseball operated as a parallel institution — its own teams, its own stadiums (often rented from MLB clubs), its own All-Star game (the East-West Classic, which routinely outdrew the MLB All-Star Game), and its own statistics, kept in newspaper box scores and league record books that would not be systematically compiled for decades.

Figure 1. Timeline of the color barrier. The gap between the leagues' end (~1958) and MLB's statistical recognition (2024) is longer than the leagues themselves operated.

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Integration: team by team

Integration did not happen all at once. It happened franchise by franchise, over 12 years, 3 months, and 6 days.

Order	Team	Date	Player(s)	League
1	Brooklyn Dodgers	April 15, 1947	Jackie Robinson	NL
2	Cleveland Indians	July 5, 1947	Larry Doby	AL
3	St. Louis Browns	July 17, 1947	Hank Thompson & Willard Brown	AL
4	New York Giants	July 8, 1949	Monte Irvin & Hank Thompson	NL
5	Boston Braves	April 18, 1950	Sam Jethroe	NL
6	Chicago White Sox	May 1, 1951	Minnie Minoso	AL
7	Philadelphia Athletics	Sept. 13, 1953	Bob Trice	AL
8	Chicago Cubs	Sept. 17, 1953	Ernie Banks & Gene Baker	NL
9	Pittsburgh Pirates	April 13, 1954	Curt Roberts	NL
10	St. Louis Cardinals	April 13, 1954	Tom Alston	NL
11	Cincinnati Reds	April 17, 1954	Nino Escalera & Chuck Harmon	NL
12	Washington Senators	Sept. 6, 1954	Carlos Paula	AL
13	New York Yankees	April 14, 1955	Elston Howard	AL
14	Philadelphia Phillies	April 22, 1957	John Kennedy	NL
15	Detroit Tigers	June 6, 1958	Ozzie Virgil Sr.	AL
16	Boston Red Sox	July 21, 1959	Pumpsie Green	AL

Three teams integrated after April 1955 — the Yankees, Tigers, and Red Sox — and the data flags them as “resistant.” The Yankees, baseball’s most visible franchise, waited eight years after Robinson’s debut. The Red Sox were last, holding out a full twelve years, three months, and six days. By the time Pumpsie Green took the field in Boston, the Negro Leagues had effectively

ceased to exist as a going concern. The Red Sox did not integrate in response to the Negro Leagues; they integrated because they were the last team in professional baseball that had not.

Figure 2. Integration timeline by franchise. Teams ordered by date of first Black player appearance.

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The pace of representation

Even after a team’s first Black player appeared, representation grew slowly. In 1947, the year of Robinson’s debut, there were 6 Black players across 544 total major-league roster spots — 1.1%. By 1956, the peak year in our dataset, 20 Black players held roster spots across 16 teams — 3.6% of the player population. The climb from tokenism to genuine representation was not a moment; it was a decade.

Figure 3. Black player representation in MLB by year, 1947-1959, as both raw count and percentage of total roster spots.

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4. The Record Book

On May 29, 2024, MLB officially merged Negro Leagues statistics into its historical record. The effect on the leaderboards was substantial: 90 total rank changes across the top-20 lists in five categories (batting average, slugging percentage, and OPS, each for career and single-season).

Career leaders

The most consequential change was at the top.

Stat	Pre-2024 #1	Value	Post-2024 #1	Value
Career AVG	Ty Cobb	.366	Piper Davis	.496
Career SLG	Babe Ruth	.690	Piper Davis	.791
Career OPS	Babe Ruth	1.164	Piper Davis	1.329

Josh Gibson, whose name is most closely associated with the record-book changes, enters at #3 in career AVG (.402), #2 in career SLG (.782), and #2 in career OPS (1.273). The single-season OPS leaderboard is even more striking: Willard Brown's 1.588 season displaces Barry Bonds's 1.422 (2004) from the top spot, with Gibson holding three of the top five slots.

The new entries are not marginal additions at the bottom of the leaderboards. Of the 16 new entrants in the career AVG top 20, the lowest-ranked enters at #20 (Roy Campanella, .351). They are distributed throughout the ranking, displacing established names at every level.

Category	Career new entries	Season new entries	Total rank changes
AVG	16	20	36
SLG	11	16	27
OPS	13	14	27
Total	40	50	90

What the leaderboards do and don't tell us

The merged leaderboards are a legitimate historical correction: these players existed, they played professional baseball at the highest level, and their statistics were recorded. Counting them is not an act of generosity; it is an act of accuracy.

But the comparison is not perfectly apples-to-apples. Negro Leagues players compiled their statistics in shorter seasons, against a different competitive pool, under different conditions (travel, equipment, field quality). The lower at-bat thresholds mean that rate statistics carry wider confidence intervals. A .496 career average over 500 at-bats is extraordinary, but it is not the same evidentiary weight as a .366 average over 11,434 at-bats.

We present both the pre-2024 and post-2024 leaderboards because the comparison itself is the finding. The magnitude of the disruption — 90 rank changes — is the point. These were not marginal players nudging into the bottom of the list. They were, by any reasonable statistical standard, among the best ever to play.

5. The Lost Seasons

This is where the cost of the barrier becomes quantifiable.

We define a “lost season” as a year in which a Negro Leagues player was active and of major-league caliber but barred from MLB by the color line. Of the 480 players in the FiveThirtyEight database, 453 had at least one lost season. The aggregate: **4,472 lost player-seasons**.

To translate seasons into a common currency, we estimate lost WAR using an OPS-based proxy calibrated against known MLB performance:

$$WAR_{est} = (OPS - lgOPS) \times AB / 600 \times 8 + 1.5$$

This formula estimates the wins above replacement a player would have produced in a full MLB season, adjusted for the league-average OPS of the era. We then apply age-curve weighting: a ramp from 0.7 to 1.0 between ages 21 and 27 (reflecting the typical development curve), a 5% annual decline after 27, and a floor at 0.3 (below which a player would likely have been released). The total:

2,063.5 estimated lost WAR.

For context: the average Hall of Famer accumulates roughly 60-70 career WAR. The aggregate lost WAR is equivalent to erasing approximately 30 Hall of Fame careers from the record. Put differently: 2,063.5 wins is more than the total franchise WAR of many MLB teams across the same period.

The men who lost the most

Player	Position	Active years	Career WAR	Lost seasons	Lost WAR	Note
Turkey Stearnes	OF	1923-1940	48.5	16	33.7	Career ended 7 years before integration. Hall of Famer.
Ray Brown	P	1931-1945	37.3	15	29.2	Career ended 2 years before integration. Hall of Famer.

Player	Position	Active years	Career WAR	Lost seasons	Lost WAR	Note
Josh Gibson	C	1930-1946	40.2	15	28.7	Died January 20, 1947 — 85 days before Robinson's debut.
Dobie Moore	SS	1920-1926	29.0	7	25.7	Entire career predates integration.
Cristobal Torriente	OF	1920-1932	30.6	13	25.5	Entire career predates integration. Hall of Famer.
Satchel Paige	P	1927-1947	39.3	17	24.7	Pitched 20 Negro Leagues seasons. Debuted in MLB at 42.
Willie Wells	SS	1924-1948	46.6	17	24.6	Played 23 seasons before integration. Hall of Famer.
Oscar Charleston	OF	1920-1941	48.9	14	24.3	Entire career predates integration. Hall of Famer.
Cool Papa Bell	OF	1922-1946	26.1	17	13.8	Career ended 1 year before integration.

Player	Position	Active years	Career WAR	Lost seasons	Lost WAR	Note
						Hall of Famer.

Turkey Stearnes lost the most WAR of any player in the dataset: 33.7 wins above replacement across 16 seasons. His career ended in 1940 — seven years before Robinson’s debut. He was elected to the Hall of Fame in 2000, sixty years after he last played. He did not live to see it.

Josh Gibson is the most iconic case. A catcher of historic ability — career WAR of 40.2, FiveThirtyEight skill ratings of 100 in both hitting and power — he played 15 seasons entirely within the color barrier. He died on January 20, 1947, at age 35. Jackie Robinson debuted on April 15, 1947. Eighty-five days separated Gibson’s death from the opening he never got.

Satchel Paige presents a different kind of loss. He did eventually reach the major leagues, but not until 1948, when he was 42 years old. He had pitched 17 full seasons in the Negro Leagues by then. His MLB career — 6 seasons, a 28-31 record, a 3.29 ERA — is the tail end of what the data suggests was a 20-year peak. He was elected to the Hall of Fame in 1971, the first player inducted primarily for his Negro Leagues career.

Figure 4. The Lost Seasons: top 20 players by estimated lost WAR. Bar length represents lost WAR; marker indicates career endpoint relative to integration date.

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What the lost-WAR estimate can and cannot tell us

The OPS-based WAR proxy is a rough instrument. It does not account for defense, baserunning, or positional value. The age-curve weighting assumes a standard aging pattern that may not fit every player. The at-bat totals from Negro Leagues seasons are small enough that individual-season WAR estimates carry substantial uncertainty.

What the estimate can tell us is the order of magnitude. Whether the true aggregate is 1,800 or 2,300, the conclusion is the same: the color barrier cost professional baseball thousands of wins worth of talent. The individual stories confirm what the aggregate suggests. These were not replacement-level players excluded from the game; they were, in many cases, the best players alive.

6. Career Projections: The Counterfactual

The lost-seasons analysis measures what the Negro Leagues players themselves lost. The career projections ask a related but different question: what did the players who *did* cross the color line lose to it?

Jackie Robinson debuted in the major leagues at age 28. Willie Mays debuted at 20. If Robinson had been born white — or born thirty years later — he would have entered the majors at 21 or 22, like any other prospect of his caliber. What would his career have looked like?

We fit quadratic WAR-by-age curves to each player's actual MLB production and extrapolate backward to age 21. The method is simple, transparent, and deliberately conservative: it assumes the player's development would have followed the trajectory implied by his actual performance, with no adjustment for the possibility that earlier development in MLB's infrastructure might have produced even better results.

Projections for players who crossed the line

Player	Position	Debut age	Lost seasons	Actual WAR	Projected lost WAR	Full career WAR
Jackie Robinson	2B	28	7	37.4	8.2	45.6
Sam Jethroe	CF	33	12	11.2	7.6	18.8
Roy Campanella	C	26	6	33.3	2.7	36.0
Elston Howard	C	26	5	39.0	2.7	41.7
Monte Irvin	OF	30	9	22.7	1.9	24.6
Larry Doby	CF	23	3	44.1	1.7	45.8

Jackie Robinson's case is the most legible. He debuted at 28, won Rookie of the Year, and produced 37.4 career WAR over ten seasons before retiring at 37. The quadratic fit suggests 8.2 additional WAR from seven lost seasons, for a projected full career of 45.6 WAR. Robinson was an MVP-caliber player from the moment he entered the league; the projection implies he would have been accumulating WAR at a comparable rate in his mid-twenties.

Roy Campanella debuted at 26 and was one of the best catchers in baseball for a decade: three MVP awards in five years. His career was cut short at 36 by a car accident that left him paralyzed. The model estimates 2.7 WAR from six lost seasons — a modest figure that reflects the conservative nature of the quadratic fit. The real loss for Campanella may be less about WAR and more about longevity: six additional seasons at the front of his career, followed by the same accident at 36, would have given him a meaningfully longer body of work.

Sam Jethroe’s case is perhaps the most striking. He debuted at 33 — an age when most players are declining — and still won Rookie of the Year. The model projects 7.6 lost WAR across twelve lost seasons. Jethroe’s peak came so late that the model cannot confidently reconstruct what his twenties would have looked like. What is clear is that a player who was the best rookie in the National League at 33 was, almost certainly, significantly better at 25.

Validation: players who debuted without delay

To test the projection method, we include three control cases: players who debuted at or near age 21 and therefore lost zero seasons to the barrier.

Player	Actual debut	Model debut	Actual WAR	Projected WAR	Residual
Willie Mays	1951 (age 20)	1952	96.6	96.6	0.0
Hank Aaron	1954 (age 20)	1955	105.7	105.7	0.0
Frank Robinson	1956 (age 20)	1956	89.3	89.3	0.0

The model correctly projects zero lost WAR for players who experienced no delay. This is a necessary but not sufficient validation — the model should produce nulls where no barrier effect exists, and it does.

Figure 5. Career WAR trajectories for Jackie Robinson and Roy Campanella, with projected pre-debut seasons shown in dashed lines.

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What the projections can and cannot tell us

The projections are counterfactual, and counterfactuals are inherently speculative. We do not know that Robinson would have produced 8.2 WAR in his twenties; we know that a quadratic curve fitted to his actual production implies it. The real Robinson might have been better (earlier access to MLB coaching and competition) or worse (the specific pressures of being the first might have sharpened his play in ways the model cannot capture). The projection is a best estimate, not a fact.

What the projections do establish is that the barrier's cost was not limited to the players who never crossed it. Robinson, Campanella, Doby, Irvin, Howard, Jethroe — every one of them entered the major leagues with their best years partially or fully behind them. The barrier did not just exclude players; it truncated the careers of those it eventually admitted.

7. The Sixty-Five-Year Gap

There is a final number worth sitting with.

The Negro Leagues operated as organized professional baseball from approximately 1920 to the late 1950s — roughly 40 years. Major League Baseball recognized their statistics as official in 2024. The gap between the end of the Negro Leagues and the recognition of their records is approximately 65 years.

The gap is longer than the thing itself.

For 65 years, the players in this dataset existed in a statistical limbo: acknowledged in the Hall of Fame (which began inducting Negro Leaguers in 1971), celebrated in retrospectives and documentaries, but absent from the official record book. Josh Gibson was a Hall of Famer whose batting average did not appear on the same leaderboard as Ty Cobb's. Satchel Paige's plaque was in Cooperstown, but his statistics were not in the Baseball Encyclopedia.

The 2024 merger corrected this — partially. The statistics are now integrated. But the 65-year gap is itself a datum. It tells us something about the velocity of institutional acknowledgment, and about how long a correction can take even when the underlying facts are not in dispute. Nobody argued that Gibson was not a great player. Nobody claimed the Negro Leagues were not real. The delay was not epistemic; it was institutional.

8. What This Is and Isn't

This is a statistical analysis of baseball's racial barrier, using publicly available data processed through a reproducible pipeline. We measure the barrier's effects on leaderboards, on individual careers, and on the aggregate talent pool. We present the numbers as precisely as the data allows and flag every point where the data's limitations constrain our conclusions.

This is not a social history of the Negro Leagues, the civil rights movement, or the lived experience of segregation. Statistics cannot capture what it meant for Josh Gibson to be the best catcher in baseball and know he would never play in the World Series. They cannot capture what it cost Jackie Robinson to be first, or what it meant for Pumpsie Green to be last. The numbers are a skeleton; the human reality is the body.

This is not an argument that the 2024 statistical merger resolves the historical injustice. It does not. Recognition that arrives 65 years late is better than no recognition, but it is not justice. The men in this dataset are dead. The audiences who watched them play are largely dead. The merger corrects a record; it does not correct a history.

What we can do — what this paper attempts — is make the shape of the barrier visible in numbers. How long it lasted (47 years). How many players it affected (453 with lost seasons). How much talent it excluded (2,063.5 estimated WAR). How slowly it fell (12 years, 3 months, 6 days from first to last team). How long recognition took (65 years). These are not the most important things about the color line in baseball, but they are the things we can measure, and measuring them precisely is a form of respect for the record.

9. Known Limitations

- **WAR estimation.** Our OPS-based WAR proxy is a simplified model. It does not account for defense, baserunning, catcher framing, or positional scarcity. The age-curve weighting assumes standard aging patterns. Individual WAR estimates may be off by 20-30%; the aggregate is more stable, but the true total could range from roughly 1,700 to 2,400.
- **Sample size.** Negro Leagues seasons produced roughly one-third the at-bats of MLB seasons. Rate statistics (AVG, SLG, OPS) computed from 80-250 at-bats carry substantially wider confidence intervals than the same statistics computed from 500-650 at-bats. The leaderboard comparisons should be read with this asymmetry in mind.

- **Survivorship bias in the data.** The 480 players in the FiveThirtyEight database are not all Negro Leagues players — they are the players whose records survived well enough to be analyzed. Players with incomplete records, or who played for teams with poor record-keeping, are underrepresented. The lost-seasons estimate of 4,472 is therefore likely a lower bound.
- **Competitive-pool differences.** Negro Leagues and MLB drew from overlapping but not identical talent pools, and the competitive environment differed in ways that affect statistical comparison. Negro Leagues teams frequently played non-league exhibition games, barnstorming tours, and winter-league ball that may or may not be captured in the statistics we use.
- **Counterfactual fragility.** The career projections in Section 6 assume that players would have followed the aging curve implied by their actual MLB performance. This assumption is strong: players who debuted late may have been unusually durable (selection effect), and earlier entry into MLB would have changed their development in unpredictable ways.
- **Confidence tiers.** Not all numbers in this paper carry equal weight. We assign three tiers:
 - **HIGH CONFIDENCE:** Official box scores from the Lahman database, MLB-ratified Seamheads statistics, integration dates from documented first appearances. These are facts.
 - **CANDIDATE:** Retrosheet game records, FiveThirtyEight player ratings, Negro Leagues WAR estimates. These are well-sourced estimates with known methodological assumptions.
 - **SPECULATIVE:** Counterfactual career projections, aggregate lost-WAR totals. These are models, not measurements. They indicate order of magnitude, not precise values.

10. Open Data and Reproducibility

The analysis pipeline consists of six Python scripts, each corresponding to one analytical stage:

Stage	Script	Function
0	00_acquire.py	Acquire and merge Lahman, Retrosheet, and 538 datasets

Stage	Script	Function
1	<code>01_leaderboards.py</code>	Compute pre- and post-merger leaderboards
2	<code>02_integration_timeline.py</code>	Build franchise-by-franchise integration timeline
3	<code>03_lost_seasons.py</code>	Calculate lost seasons and lost WAR for Negro Leagues players
4	<code>04_career_projections.py</code>	Fit WAR-by-age curves and project counterfactual careers
5	<code>05_export.py</code>	Export derived datasets for dashboard consumption

All code is available at [repository URL]. The pipeline is deterministic: given the same input data, it produces identical outputs. Source data (Lahman, Retrosheet, FiveThirtyEight) is publicly available from the respective projects.

The interactive dashboard allows readers to explore the merged leaderboards, step through the integration timeline, examine individual player profiles, and test sensitivity of the lost-WAR estimates to alternative age-curve assumptions.

11. Methodology Notes

WAR estimation formula

For Negro Leagues players without official WAR calculations, we estimate WAR from available offensive statistics:

$$WAR_{est} = (OPS - lgOPS) \times (AB / 600) \times 8 + 1.5$$

Where `lgOPS` is the league-average OPS for the relevant season. The multiplier of 8 scales the OPS differential to approximate the run-to-win conversion used in standard WAR calculations. The constant of 1.5 represents the approximate value of a replacement-level player. This formula is a deliberate simplification; we use it because more sophisticated WAR models require defensive and baserunning data that is not consistently available for Negro Leagues players.

Age-curve weighting

For lost-seasons calculations, we weight each lost season by an age-based factor:

- Ages 21-27: linear ramp from 0.7 to 1.0 (development years)
- Ages 27+: 5% decline per year (standard aging curve)
- Floor: 0.3 (below which a player would likely not hold a roster spot)

This curve is derived from the empirical aging patterns observed in MLB data and represents a consensus estimate rather than a precise individual prediction.

Career projection model

Counterfactual careers are projected using a quadratic (degree-2 polynomial) fit to the player's actual WAR-by-age data, extrapolated backward to age 21. Projected WAR values below zero are floored at zero. The quadratic form was chosen for interpretability and stability; higher-degree polynomials produced unreliable extrapolations, particularly for players with short MLB careers.

Acknowledgments

Data sourced from the Lahman Baseball Database (Sean Lahman), Retrosheet (retrosheet.org), FiveThirtyEight (fivethirtyeight.com), and the Seamheads Negro Leagues Database (seamheads.com). Library of Congress photographs used under public domain. Integration dates compiled from Baseball Reference and primary historical sources.

The greatest players in the history of the game were not allowed to play in it. The numbers were always there. We just were not counting them.

Any errors of methodology, interpretation, or framing are the author's alone.

Report 03 of 10 in the One Hundred Years series. colorline.onehundredyears.report

Contact: hello@onehundredyears.report **License:** MIT **Repository:** github.com/doctorbrownphd/hundred-years-baseball